

Christian Lezcano

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Data Engineer | Machine Learning Engineer

Data Engineer with a specialization in **developing scalable data infrastructure and implementing production-ready AI solutions**. My Core technical expertise lies in **Python, SQL, PostgreSQL, and AWS**, complemented by strong capabilities in **Machine Learning** utilizing frameworks like **TensorFlow and PyTorch**. I have applied these skills across diverse sectors, including beverage manufacturing, higher education, and government. My technical toolkit also includes proficiency in **ETL processes (DBT), data visualization (Power BI, MetaBase), MLOps (MLflow), containerization (Docker), CI/CD (GitHub Actions), and cloud data warehousing (Amazon Redshift)**, along with libraries like **Keras and Hugging Face**. I am seeking a challenging opportunity where I can leverage this comprehensive skill set to design robust data pipelines, build impactful predictive models, and contribute significantly to data-driven innovation within a forward-thinking organization.

Work Experience

Data Engineer at Vitaly Mineral Water

Sep 2016 to date

- Built a RAG system for supply chain optimization, integrating LangChain, GPT-4, Llama 2, and conversational AI. Developed an ELT pipeline with Pandas, Airflow, PostgreSQL, and AWS S3, leveraging NLTK, SentenceTransformers, and FAISS for semantic search. Optimized with FastAPI and Redis caching, reducing disruptions by 20%. Deployed on AWS with Docker, Prometheus, and Grafana for robust, real-time monitoring.
- Developed scalable microservices using Python and AWS (S3, RDS PostgreSQL, EC2), improving data processing and integration across multiple production facilities, resulting in a 30% reduction in data latency and enhanced system scalability.
- Built and optimized PostgreSQL databases, implementing indexing strategies and writing complex SQL queries, leading to a 40% improvement in query performance and faster data retrieval for critical business operations.
- Designed and deployed predictive models for demand forecasting and maintenance scheduling using Python and scikit-learn, achieving a 20% increase in operational efficiency through accurate regression analysis.
- Implemented CI/CD pipelines using GitHub Actions to streamline code deployment and ensure high code quality, reducing deployment time by 50% and minimizing code-related issues.
- I took ownership of our analytical data workflows. This involved integrating and preparing data from diverse sources using advanced SQL, and then transforming it into high-impact, automated reports and visualizations with Excel/Power BI, boosted by DAX and VBA for deeper insights and efficiency.

Main Technologies: *Python , SQL , PostgreSQL , Microsoft Power BI , MetaBase , Microsoft Excel , Microsoft SQL Server , Extract, Transform, Load (ETL) , Data Build Tool (DBT) , Machine Learning , Apache NiFi , Docker , scikit-learn , TensorFlow , PyTorch , Keras , MLflow , Amazon Web Services (AWS) , RAG , LangChain, FAISS , Amazon Redshift , Microsoft Excel (VBA, Power Query, Advanced Formulas), Microsoft Power BI (DAX) , Pentaho.*

AI and Data Science Academic Instructor at Columbia University of Paraguay

Jul 2023 to date

- Advise undergraduate students on AI and data science projects, guiding them through the design and implementation phases, resulting in successful project completions and innovative research outcomes.
- Supervise students' progress, providing regular technical feedback and personalized support to overcome complex challenges, leading to a 20% improvement in project quality and timely submissions.
- Foster independent problem-solving and critical thinking, encouraging students to apply theoretical concepts to practical scenarios, enhancing their technical competence and confidence in AI-related projects.

Main Technologies: *Python , TensorFlow, PyTorch, Keras, scikit-learn , Pandas , NumPy , Hugging Face , Jupyter Notebooks , University Teaching , Research and Development (R&D) , Academic Research.*

Data Research Analyst at Columbia University of Paraguay

Oct 2023 to Oct 2024

Led a research-driven project to enhance academic efficiency, initially for the Computer Science program. Innovatively proposed a suite of new Key Performance Indicators (KPIs)—covering critical areas such as enrollment, retention, graduation, and academic performance—and developed a comprehensive analytical framework. This scalable system provides novel metrics and methodologies for data-driven insights into student success, designed for potential university-wide application to foster continuous improvement.

Main Technologies: *Key Performance Indicators , Research and Development (R&D) , Academic Research.*

Machine Learning & AI Projects

Financial Advisory RAG Chatbot

- Built a scalable RAG-powered chatbot leveraging NASDAQ company documents, integrating LangChain with OpenAI GPT to deliver accurate financial insights through conversational AI, improving data accessibility for non-technical users.
- Engineered an end-to-end data pipeline using PyPDF2, NLTK/spaCy, custom chunking algorithms, and FAISS indexing, enabling efficient ingestion and retrieval of thousands of financial documents.
- Optimized chatbot performance by developing a FastAPI/Uvicorn backend with Redis caching and asyncio, reducing response time latency by over 50% and ensuring real-time interactions.
- Deployed the production system with Docker and AWS S3, implementing Prometheus monitoring for system reliability and performance tracking, while achieving enhanced accuracy through hybrid BM25 + vector search with RAGAS-based validation.

Main Technologies: *Python, LangChain, OpenAI GPT, FastAPI, Docker, AWS S3, FAISS, Redis, Prometheus, Streamlit, RAGAS*

Skills

Tech Skills: *Python , SQL , PostgreSQL , Microsoft Power BI , Microsoft Excel , Microsoft Power BI (DAX) , Microsoft SQL Server , Extract, Transform, Load (ETL), Machine Learning , Apache NiFi , TensorFlow , PyTorch , Data Build Tool (DBT) , Redis , Docker , MetaBase , Pentaho , Amazon Web Services (AWS), Amazon Redshift , scikit-learn , MLflow , TensorFlow , PyTorch , Keras , Hugging Face , LangChain, OpenAI GPT , FastAPI , FAISS , Redis , Prometheus , Streamlit , RAGAS.*

Agile Methodologies: *Scrum, Kanban*

Education

Anyone AI

Specialization in machine Learning Engineering 2025

National University of Asuncion

M.S., Computer Science 2012

National University of Asuncion

B.S., Computer Science Engineering 2008

Certifications

IBM Data Science Professional Certificate

2023

Publications

- Lezcano, C. and Arias, M. (2019b), "Synthetic dataset generation with itemset-based generative models", in *Proceedings of the 4th Workshop on Reliability and Security Data Analysis co-located with IEEE International Symposium on Software Reliability Engineering (ISSRE 2019)*, Berlin, Germany, October 28th, 2019., pp. 288-293.
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- Lezcano, C. and Arias, M. (2019a), "Characterizing transactional databases for frequent itemset mining", in *Proceedings of the 1st Workshop on Evaluation and Experimental Design in Data Mining and Machine Learning co-located with SIAM International Conference on Data Mining (SDM 2019)*, Calgary, Alberta, Canada, May 4th, 2019., pp. 44-53.
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- Colbes, J. D., Corona, R. I., Lezcano, C., Rodriguez, D. O. and Brizuela, C. A. (2017), "Protein side-chain packing problem: is there still room for improvement?", *Briefings in Bioinformatics* 18(6), 1033-1043.
[\[pdf\]](#) [\[bibtex\]](#)
- Brizuela, C. A., Corona, R. I., Lezcano, C., Rodriguez, D. O. and Colbes, J. D. (2015), "An experimental analysis of the performance of sidechain packing algorithms", in *Genetic and Evolutionary Computation Conference, GECCO 2015*, Madrid, Spain, July 11-15, 2015, Companion Material Proceedings, pp. 929-933.
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- Lezcano, C., Pinto, D. and Baran, B. (2008), "Team algorithms based on ant colony optimization - A new multi-objective optimization approach", in *Parallel Problem Solving from Nature - PPSN X*, 10th International Conference Dortmund, Germany, September 13-17, 2008, Proceedings, pp. 773-783.
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